Short Selling and Readability in Financial Disclosures: A Controlled Experiment

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Keywords: Regulation SHO; Short-selling; Annual report readability; Tone ambiguity; Limited attention

JEL Codes: G14, G18, M41

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Abstract

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1 INTRODUCTION

An important channel for corporate managers to communicate a firm's financial disclosures to investors and analysts is the Form 10-K (annual report). Market participants and regulators care about the quantity of information available to the public and the quality of the information provided in financial reports. Consequently, the existing research develops annual report readability measures to assess the effective communication of valuation-relevant information between the firm and capital market participants (Loughran & McDonald, 2014; Bonsall et al., 2017). Recent research connects annual report readability to the obfuscation of earning-relevant information (Li, 2008; Bushee, Gow, & Taylor, 2018), borrowing costs (Ertugrul et al., 2017), the valuation of closed-end funds (Hwang & Kim, 2017), management risk incentive (Chakrabarty et al., 2018), the cost of equity (Rjiba et al., 2021), and employee wages (Bai, Serfling, & Shaikh, 2022).

This paper investigates the causal effect of short-sale constraints on annual report readability. Recent research suggests that short sellers can identify financial misconduct (Karpoff & Lou, 2010) and help to improve the quality of financial reporting by significantly reducing earnings management (Massa, Zhang, & Zhang, 2015; Fang, Huang, & Karpoff, 2016). Yet, when encountering short-selling pressure, how firms present valuation-relevant information to investors is unknown. Investigating how short selling affects annual report readability can provide insight into management decisions on another dimension of information disclosure quality.

Testing the causal effect of short selling on annual report readability is challenging due to its endogenous nature. Our identification strategy is based on Regulation SHO (hereafter Reg SHO), which removes short-sale price tests for 986 pilot firms that are randomly chosen from the

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Russell 3000 Index between May 2, 2005 and August 6, 2007. Prior studies document that short-selling volume has been increased considerably for pilot stocks as opposed to non-pilot stocks (SEC 2007; Diether, Lee, & Werner, 2009; Grullon, Michenaud, & Weston, 2015). Consequently, we can examine the causal effect of short-sale constraints on annual report readability using a difference-in-differences (hereafter, DiD) method.

We begin by confirming that pilot stocks are randomly chosen by comparing the firm characteristics of the pilot and non-pilot firms one year before the announcement of Reg SHO. We use the Bog Index (Bonsall et al., 2017) as a proxy for annual report readability. The index is based on the computational linguistics software program *StyleWriter – The Plain English Editor* and captures the SEC Plain English writing principles including jargon, passive voice, weak verbs, overused and complex words, and sentence length (SEC 1998). An annual report with a higher Bog index is considered less readable.

Our analysis reveals that, before Reg SHO, pilot and non-pilot firms display similar characteristics, and our pre-trend analysis confirms a parallel trend assumption. We then run DiD regression analysis to investigate how the relaxation of short-sale constraints impacts annual report readability. We demonstrate that readability for the pilot firms is significantly lower than that for the non-pilot firms during the Reg SHO compared to the pre-Reg SHO period. Economically, pilot firms produce a 7.3% of a standard deviation lower annual report readability than non-pilot firms during the pilot program period compared to the pre-event period.

Our results accord well with Bloomfield's (2002) incomplete revelation hypothesis, which predicts that investors with limited attention have less trading interests in firms with harder-to-

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process information, resulting in a slower incorporation of valuation-relevant information into stock prices. When short-sale constraints are less rigorous, pilot firms bury mandated earnings-relevant information in less readable 10-Ks, making it more costly for investors to comprehend. This strategy helps to reduce the potential amount of short-selling for three reasons. First, investors have limited attention and information processing power (Fang, Peress, & Zheng, 2014). Second, investors place less weight on, pay less attention to, and even neglect complex and hard-to-process information (Hirshleifer & Teoh, 2003; Peng & Xiong, 2006; Cohen & Lou, 2012). Third, firms exploit the limited attention of investors (Daniel, Hirshleifer, & Teoh, 2002).

If the above argument holds true, firms that receive less investor attention are more likely to manipulate annual report readability when faced with short-selling pressure. Using abnormal trading volume and extreme returns (Barber & Odean, 2008), and analyst coverage (Hirshleifer & Teoh, 2003) as proxies for investor attention, we re-estimate DiD regressions using high and low attention subsamples. We demonstrate that the effect of short selling on annual report readability is more pronounced for firms with lower abnormal trading volume, extreme returns, and analyst coverage.

Firms with poor growth prospects are more likely to strategically adjust their disclosure in response to Reg SHO. We employ four measures of firms' prospects, including the cash-flow-to-price ratio, earnings-to-price ratio, and book-to-market ratio (Dechow et al., 2001), and Tobin's Q (Lang, Ofek, & Stulz, 1996), and conduct DiD regressions using poor and strong growth prospect subsamples. We find that the effect of Reg SHO on annual report readability is more pronounced among firms with poorer growth prospects.

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Ambiguous text in 10-Ks can also obstruct investors' ability to comprehend reports. Loughran and McDonald (2011) link higher uncertainty word usage in annual reports to increased stock return volatility post-10-K filing. Loughran and McDonald (2013) show that IPOs with more uncertainty words display greater absolute offer price revisions, higher first-day returns and subsequent return volatilities. Furthermore, Ertugrul et al. (2017) show that firms using more uncertainty words in 10-Ks face higher borrowing costs. We demonstrate that pilot firms use greater uncertainty text in their annual reports during the Reg SHO periods.

Our study makes several contributions to the literature. First, we expand on the impacts of short-selling on financial reporting quality (Massa, Zhang, & Zhang, 2015; Li & Zhang, 2015; Fang, Huang, & Karpoff, 2016) by investigating its causal impact on readability and tone ambiguity in 10-K reports. Second, we find that short-sale constraints can be an important determinant of readability and tone ambiguity in 10-Ks, complementing the literature on linguistic features of corporate disclosures (Li, 2008; Bloomfield, 2008; Loughran & McDonald, 2016). Third, our paper complements the growing literature on the impact of annual report readability on the financial market.[1] Finally, our study complements the debates on the costs and benefits of short selling (Haruvy & Noussair, 2006; Goldstein & Guembel, 2008). We show that the relaxation of short-sale constraints leads firms to produce less transparent 10-Ks.

[1] Researchers have examined the impact of annual report readability on earnings persistence (Li, 2008), the trading activities of small and retail investors (Miller, 2010; Lawrence, 2013), return volatility and earnings forecast errors and dispersion (Loughran & McDonald, 2014; Bonsall et al., 2017), borrowing costs (Ertugrul et al., 2017), the valuation of closed-end funds (Hwang & Kim, 2017), employee wages (Bai, Serfling, & Shaikh, 2022), capital investment efficiency (Biddle, Hilary, & Verdi, 2009), analyst coverage and analyst dispersion (Lehavy, Li, & Merkley, 2011), management risk incentive (Chakrabarty et al., 2018)), the cost of equity (Rjiba et al., 2021), and litigation risk (Humphery-Jenner et al., 2022). For a detailed review, see Loughran and McDonald (2016).

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Our paper is related to Li and Zhang (2015), who were the first to link short selling to annual report readability. They find that in response to Reg SHO, pilot firms with bad news file less readable annual reports. They suggest that "there is an ongoing debate over the appropriate measurement of financial disclosure readability. Therefore, our analysis on annual report readability is explorative in nature, and we leave an in-depth examination of changes in financial report readability around Reg SHO to future research".

Our paper complements their study in several ways. First, Li and Zhang (2015) focus on the effect of Reg SHO on firms with bad news. It is unclear whether short selling affects 10-K readability for all firms. We fill this gap by establishing a causal link between short selling and 10-K readability for all firms under Reg SHO. Second, Li and Zhang (2015) use the Gunning (1952) Fog Index, which has been critiqued for measuring 10-K readability inadequately (Loughran & McDonald, 2014). In contrast, we show consistent results using the Bog index, a novel annual report readability measure. Third, we introduce new evidence that investor attention and firms' prospects drive the adverse effect of Reg SHO on 10-K readability. Finally, our paper provides new evidence that Reg SHO significantly impacts the tone ambiguity of annual reports.

The rest of the paper is organized as follows. Section 2 discusses related literature and hypothesis development. Section 3 describes the data and presents summary statistics. The main findings and robustness tests are discussed in Sections 4 and 5. We provide concluding remarks in Section 6.

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2 RELATED LITERATURE AND HYPOTHESES DEVELOPMENT

We discuss the literature on how short selling can affect financial markets and corporate decisions in Section 2.1. The studies of readability and tone ambiguity in financial disclosures are described in Section 2.2. We intergrate the two strands of literature and develop our hypotheses in Section 2.3.

2.1 Regulation SHO

In July 2004, to evaluate short-selling restrictions' effectiveness and necessity, the SEC announced Reg SHO which removed the uptick rule for 986 randomly selected stocks (pilot stocks) from the Russell 3000 Index between May 2, 2005 to August 6, 2007.[2] Researchers find that although short-selling volume has been shown to increase for pilot firms, there is no significant difference in market quality between pilot and non-pilot firms (SEC 2007, Alexander & Peterson, 2008; Diether, Lee, & Werner, 2009; Grullon, Michenaud, & Weston, 2015). The SEC eventually removed the uptick rule for all public firms on July 6, 2007. The suspension of the uptick rule for all firms drew critiques from corporate managers and former regulators, especifically during the 2008 global financial crisis. On February 24, 2010, the SEC imposed a short-sale restriction if a stock price tanked by 10% or more from the previous day's closing price.

There is a rich literature on how short selling impacts asset prices (Jones & Lamont, 2002; Boehmer, Jones, & Zhang, 2008; Chu, Hirshleifer, & Ma, 2020; Han et al., 2021). More recently,

[2] The uptick rule is a trading restriction imposed in 1938. Under this rule, shorting a stock is permitted only on an uptick.

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empirical studies on the effect of short selling on corporate decisions are growing. Using Regulation SHO, Grullon, Michenaud, and Weston (2015) find that the relaxation of short-sale constraints reduces investment and stock issues. Fang, Huang, and Karpoff (2016) document that short-selling activities reduce earnings management. Using Regulation SHO, others link short selling with corporate innovation (He, Tian, & Xiao, 2023), the design of executive incentive contracts (De Angelis, Grullon, & Michenaud, 2017), corporate social responsibility (Gao, He, & Wu, 2022), and bad news management forecasts and 10-K readability (Li & Zhang, 2015), workplace safety (Bai, Lee, & Chi, 2020), stock price crash risk (Deng, Kim, & Gao, 2020), audit fees (Hope, Hu, & Zhao, 2016), Bad news disclosure (Clinch, Li, & Zhang, 2019), price informativeness during and after earnings announcement (Clinch & Li, 2021), real earnings management (Cai, Guo, & Tan, 2023), corporate short-termism (Chang et al., 2021), long-run management forecasts (Chen et al., 2020) and insiders' opportunistic trading (Wang et al., 2022).

2.2 Strategic disclosure behavior: readability and tone ambiguity in financial disclosures

The existing literature documents that firms can strategically make disclosure choices to impact stock prices (See Healy & Palepu, 2001; Beyer et al., 2010, for a detailed review). For example, managers can strategically time their disclosures by withholding bad news (Kothari, Shu, & Wysocki, 2009) and expanding their disclosures to increase institutional ownership (Healy, Hutton, & Palepu, 1999) and analyst coverage (Lang & Lundholm, 1993). Furthermore, Hirshleifer and Teoh (2003) model firms' disclosure behaviors, such as pro forma earnings disclosures, in the context of limited investor attention.

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Recent literature has focused on a crucial aspect of strategic disclosure: annual report readability. Li (2008) is the first to link annual report readability with firm performance using the Guning (1952) Fog Index and find that managers can strategically use hard-to-read financial reports to obfuscate a firms' current and future performance, especially when earnings are lower. Recent evidence reveals that firms with less readable 10-K filings are associated with lower capital investment efficiency (Biddle, Hilary, & Verdi, 2009), reduced trading volume around the 10-K filing date (Miller, 2010), higher earnings forecast accuracy and decreased analyst dispersion (Lehavy, Li, & Merkleym, 2011), an increase in managerial forecasts of sales, cash flows, and earnings (Guay, Samuels, & Taylor, 2016), higher costs of borrowing (Ertugrul et al., 2016) and equity financing (Rjiba et al., 2021), greater management incentives (Chakrabarty et al., 2018), reduced litigation risk (Humphery-Jenner et al., 2022), higher employee wages (Bai, Serfling, & Shaikh, 2022), and lower valuations for equity closed-end funds (Hwang & Kim, 2016). Additionally, Kubick and Lockhart (2016) show that firms farther from the SEC are associated with greater crash risk and this result is stronger for less readable firms.

Loughran and McDonald (2014) stress that the Fog index is a poor readability measure because the percentage of complex words with three or more syllables consist of measurement errors.[3] The Fog index also lacks explanatory power for earnings surprises and analyst dispersion. Instead, they propose using 10-K file size as a new readability measure, showing that larger file sizes correlate with greater subsequent stock return volatility, analyst dispersion, and earnings

[3] For example, 52 complex words like company, corporation, operations, and management, are widely and frequently used in 10-Ks and represent more than 25% of the complex word count. However, these common financial terms are easy for investors to comprehend.

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surprises. However, Bonsall et al. (2017) argue that 10-K file size as a readability measure introduces measurement errors stemming from content unrelated to the text in 10-Ks.[4] For this reason, they develop a novel readability measure, the Bog index, which is based on the computational linguistics software program *StyleWriter – The Plain English Editor*. The index captures the SEC Plain English writing principles including complex words, sentence length, jargon, passive voice, weak verbs, and overused words. Using several controlled experiments and a readability-related regulatory intervention, they document that the Bog index is a more compelling measure of 10-K readability than other measures. Consequently, we investigate the impact of short selling on financial report readability using the Bog index.

Tone ambiguity in financial reporting is a new aspect of disclosure. Loughran and McDonald (2011) develop a list of ambiguous tones in annual reports and find that firms using fewer uncertainty words have a more positive market reaction after the 10-K filing period. Loughran and McDonald (2013) document that IPOs with higher frequencies of uncertainty words usage in Form S-1 filings exihibit greater first-day returns, higher absolute offer price revisions, and higher subsequent volatilities. Furthermore, Ertugrul et al. (2016) demonstrate that firms with a more ambiguous tone in annual reports experience higher borrowing costs.

2.3 Hypothesis development

[4] Specifically, they argue that file size consists of exhibits such as compensation contracts, or bond indentures that are disparate toward the 10-K reporting requirements and contents that are unrelated to the text in 10-Ks such as XML, HTML, PDFs. Additionally, Cahan et al. (2021) find that 10-K file size is a poor measure of readability after the adoption of XBRL.

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In this subsection, we integrate the two strands of literature to develop our hypotheses. We examine how pilot firms respond to short-selling pressure resulting from Regulation SHO in comparison to non-pilot firms. Our hypotheses center on annual report readability, investor attention, growth prospects, and the use of ambiguous language.

Under the incomplete revelation hypothesis of Bloomfield (2002), investors with limited attention show less interest in trading firms whose information is more costly to collect. Consequently, valuation-relevant information is slowly incorporated into stock prices. Pilot firms, which are easier to sell short, have an incentive to bury mandated earnings-relevant information in less readable 10-Ks that are more challenging for investors to comprehend. They adopt this strategy to potentially reduce the level of short selling, driven by three key reasons. Firstly, both novice and sophiscated investors encounter constraints in attention and information processing capabilities (Fang, Peress, & Zheng, 2014). Secondly, investors tend to assign less weight, pay reduced attention to, and sometimes neglect complex and hard-to-process information (Hirshleifer & Teoh, 2003; Peng & Xiong, 2006; Hirshleifer, Lim, & Teoh, 2011; Cohen & Lou, 2012; Davydov & Peltomäki, 2023). Thirdly, firms capitalize on the limited attention of investors (Daniel, Hirshleifer, & Teoh, 2002). This leads us to formulate our first and second hypotheses:

Hypothesis 1: Pilot firms produce less readable 10-Ks than non-pilot firms during Reg SHO periods.

Hypothesis 2: The impact of Reg SHO on annual report readability is more pronounced for firms with limited investor attention.

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Furthermore, short sellers typically target firms with poor prospects (Dechow et al., 2001). As a result, firms with poor growth prospects are more likely to strategically manage their disclosure behaviors when faced with increased short selling pressure in an effort to reduce the potential amount of short selling. This view leads to our third hypothesis:

Hypothesis 3: The effect of Reg SHO on annual report readability is more pronounced for firms with poor growth prospects.

Lastly, firms may employ ambiguous language in annual reports to impede investors' comprehension. Consequently, we propose our fourth hypothesis:

Hypothesis 4: Pilot firms use more ambiguous tones in their 10-Ks than non-pilot firms during Reg SHO periods.

3. DATA

In this subsection, we describe our sample and key variables and provide the summary statistics of these variables.

3.1 Sample

Our sample is constructed based on the Russell 3000 Index in June 2004. Our initial sample includes 2,952 stocks (986 pilot and 1,966 non-pilot stocks).[5] We choose common stocks that are

[5] Our sample of pilot and non-pilot stocks is identical to that of Fang, Huang, and Karpoff (2016). We thank Vivan Fang for sharing the Russell 3000 index merged with CRSP PERMNO numbers and FTSE Russell for sharing the Russell 3000 membership list with us.

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listed in NYSE, AMEX and NASDAQ. We include all 10-K filings (i.e., 10-K 405, 10-KSB, and 10-KSB40). We require that firms have a Compustat Permanent Company Identifier match, be ordinary common stock, have at least 2,000 words in the 10-K, and have a gap of at least 180 days between two filings.

Our control variables are from several sources. First, we collect accounting information from the CRSP/Compustat Merged Database, stock returns, and trading volume from CRSP. Second, we gather analyst coverage data from IBES and corporate events information from Thomson Reuters SDC Platinum M&A and Global New Issues databases.

Reg SHO program was implemented from May 2, 2005 to July 6, 2007. Using firms' fiscal year ending dates, we classify January 1, 2001 to December 31, 2003 as the *pre-event* period, January 1, 2005 to December 31, 2007 as the *during-event* period. We require that firms have non-missing data for all key variables. Following Fang, Huang, and Karpoff (2016), we exclude firms with fiscal year ending dates in 2004. Our unbalanced sample includes 2,831 stocks (921 pilot and 1,910 non-pilot firms). We also construct a balanced sample by requiring firms to be in the sample over the pre-event and the during-event periods. The balanced sample contains 2,550 firms (839 pilot and 1,711 non-pilot firms). We use the balanced sample for most of our tests and verify the robustness of our analysis using the unbalanced sample.

3.2 Key variables

Following Bonsall et al. (2017), we measure the annual report readability of our sample firms using the Bog index. We use the proportion of uncertainty words such as *approximate*,

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assume, *contingent*, *depend*, and *uncertain* in Loughran and McDonald (2011), to capture tone ambiguity in 10-Ks.

Following Li (2008), we control for firm characteristics that determine annual report readability. Our control variables include the natural logarithm of the market value of equity at the end of the fiscal year, the natural logarithm of the book-to-market ratio, the natural logarithm of firm age since its first appearance in the CRSP monthly return file, special items (special items to asset ratio), stock return volatility, earnings volatility, business complexity (the natural logarithm of the number of business and geographic segments), financial complexity (the natural logarithm of the number of non-missing items in Compustat), corporate events (SEO and M&A dummy variables), and Delaware incorporation dummy variable. We also include profitability (ROA) because firms with lower profitability are more likely to obscure valuation-relevant information in annual reports. Descriptions of all the variables are in the Appendix.

3.3 Summary statistics

Table 1 reports the summary statistics of all key variables from the balanced sample. All variables are winsorized at the 1% and 99% levels to minimize the influence of outliers. On average, a firm's annual report has a Bog Index of 82.95 and contains 1.37% uncertainty words. The average firm also has a natural log firm size of 7, a natural log book-to-market ratio of -0.82, a log firm age of 2.82, a log business segment of 1.06 and a log geographic segment of 1.08, return volatility of 0.12, and an earnings volatility of 0.05. Additionally, on average, a firm has a ROA of 0.03 and a special item ratio of -0.01.

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< Table 1 >

4. RESULTS

The section discusses our empirical results. First, we compare the firm characteristics of the pilot and non-pilot firms one year before the announcement of Reg SHO. Second, we investigate the effect of short-sale constraints on annual report readability in a DiD framework. Third, we examine how investor attention affects the relationship between Reg SHO and readability. Lastly, we test whether Reg SHO impacts tone ambiguity in 10-Ks.

4.1 Firm characteristics before Regulation SHO

To verify that the selection of pilot firms from the Russell 3000 index is random, we compare the firm characteristics of the pilot and non-pilot firms one year before the announcement of the program. Table 2 presents the summary statistics and mean differences of firm characteristics between the pilot (treatment) and non-pilot (control) groups for the balanced sample. We report the *t*-statistics of the two-sample *t*-tests. We find that the groups have similar firm characteristics despite pilot firms exhibiting lower earnings volatility. The results in Table 2 show that Reg SHO is a compelling experiment that is appropriate for testing the causal impact of the relaxation of short-sale constraints on readability in 10-Ks.

< Table 2 >

4.2 Difference-in-differences results

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This subsection examines the effect of short selling on annual report readability using a DiD methodology for multivariate regressions. To test Hypothesis 1, we estimate the following specification for the balanced sample:

$$Bog_{i,t} = \alpha + \beta_1 * Pilot_i + \beta_2 * Pilot_i * During_t + \beta_3 * During_t + \gamma \cdot Controls + \varepsilon_{i,t,} \quad (1)$$

where $Bog_{i,t}$ is the Bog Index for firm *i* in year *t*. $Pilot_i$ is a dummy variable that equals one if a stock is selected as a pilot stock in Reg SHO's pilot program and zero otherwise. $During_t$ is a dummy variable that equals one if the end of a firm's fiscal year *t* falls between January 1, 2005 and December 31, 2007 and zero otherwise. Following Li (2008), we control for previously shown to determine the annual report readability: size, book-to-market ratio, firm age, special items to asset ratio, stock return and earnings volatility, business complexity, financial complexity, ROA, corporate events (SEO and M&A dummy variables) and a Delaware incorporation dummy variable. In column (2), we add the industry fixed effects (3-digit SIC codes) and fiscal year fixed effects to our Equation (1). The variable $During_t$ is omitted because it is perfectly correlated with the fiscal year fixed effects. We also replace the industry fixed effect with the firm fixed effect to our regression in column (3). Thus, $Pilot_i$ is dropped because it is perfectly correlated with the firm fixed effects. All standard errors are clustered by firm.

< Table 3 >

Table 3 reports the regression results of Equation (1). The coefficient of interest is β_2, which captures the causal effect of short selling on annual report readability. Hypothesis 1 predicts

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a positive and significant coefficient (β_2). In column (1), the coefficient of $Pilot_i * During_t$, β_2 is 0.460 and is significant at the 5% level, implying that the Bog index of pilot firms is 0.460 higher than those of non-pilot firms during the Reg SHO period as opposed to the pre-Reg SHO period. Additionally, the coefficient of $Pilot_i$ is statistically insignificant, suggesting that all firms exhibit a similar Bog Index before the pilot program. This result indicates that the parallel trend assumption in a DiD regression framwork is likely to hold. We will test the parallel trend assumption in Section 5.1.

In column (2), we control for the industry and year fixed effects. Consistent with Hypothesis 1, we show that the coefficient on $Pilot_i * During_t$ is 0.458 and significant at the 5% level. To further alleviate potential omitted variable bias from unobserved firm characteristics persistent over time, we employ firm fixed effect in column (3). The variables $Pilot_i$ and $Delaware_i$ are omitted due to collinearity. The slope of $Pilot_i * During_t$ is 0.508 and significant at the 1% level. This indicates that pilot firms produce a 7.3% (0.508/6.99) of a standard deviation lower annual report readability than non-pilot firms during the pilot program period compared to the pre-event period. This difference is economically significant. When considering one of the significant existing drivers of the Bog index, the dummy variable for mergers and acquisitions (MA) events, we find that the Bog index differs by 0.239 between firms with MA activities and those without. This difference represents 3% (0.239/6.99) of a standard deviation. In sum, the evidence in Table 3 shows that pilot firms produce significantly less readable annual reports during the Reg SHO period compared to the pre-Reg SHO period.

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4.3 Cross-sectional analyzes based on investor attention

This section conducts empirical tests to validate Hypothesis 2, which posits that firms receiving less investor attention are more likely to manipulate annual report readability when facing short-selling pressure.

We use abnormal trading volume (Barber & Odean, 2008), extreme returns (Barber & Odean, 2008), and analyst coverage (Hirshleifer & Teoh, 2003) as proxies for investor attention. Stocks with lower abnormal trading volume, extreme returns and less analyst coverage are regarded as stocks that receive less attention. For each measure, we partition the samples into high and low investor attention subsamples based on their median values each year. We then repeat the analysis above for Table 3 and report the results in Table 4. We present the results with firm and fiscal year fixed effects.

< Table 4>

In Table 4, we find evidence supporting Hypothesis 2, indicating that the impact of Reg SHO on annual report readability is more pronounced in stocks with lower levels of investor attention. We find that the slope of $Pilot_i * During_t$ is 0.505 and significant for the low abnormal trading volume subsample, while the slope becomes statistically insignificant in the high abnormal trading volume subsample. Similarly, the effect of Reg SHO on readability is stronger in the low extreme returns subsample than in the high extreme returns subsample. For analyst coverage, we find that the slope of $Pilot_i * During_t$ is 0.804 and significant at the 1% level for the low coverage group, whereas it is insignificant in the high coverage.

The evidence in Table 4 shows that the effect of short-selling on annual report readability

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is significant only for stocks that receive low investor attention. Collectively, these results are in line with the incomplete revelation hypothesis for the effect of short-selling on annual report readability.

4.4 Cross-sectional analyzes based on firms' growth prospects

Firms with poor prospects are more likely to be targeted by short sellers (Dechow et al., 2001). Pilot firms face increased short-sale pressure under Regulation SHO due to relaxed constraints. Consequently, pilot firms with poor prospects are more likely to be targeted by short sellers and have stronger incentives to obscure their financial reports in response to Reg SHO. We test Hypothesis 3 in this subsection.

We employ four measures of firms' growth prospects: fundamentals-to-market ratios, including cash-flow-to-price ratio, earnings-to-price ratio, and book-to-market ratio (Dechow et al., 2001); and Tobin's Q (Lang, Ofek, & Stulz, 1996). Firms with high fundamentals-to-market ratios or low Tobin's Q are considered to have poor growth prospects. We partition the samples into subsamples of poor and strong growth prospects based on the median value of each proxy every year. Then, we re-estimate our DiD regressions with firm and year fixed effects for each subsample. The results of the DiD regressions are reported in Table 5.

< Table 5>

In line with Hypothesis 3, in each test, $Pilot_i * During_t$ is positive and significant in the subsamples with poor growth prospects (high fundamentals-to-market ratios or low Q), while it

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remains insignificant in the subsamples with strong growth prospects (low fundamentals-to-market ratios or high Q). For example, in Column (3), the coefficient of $Pilot_i * During_t$ is 0.794 and significant at the 1% level for the poor growth prospect subsample. In Column (4), it reduces to 0.274 and becomes statistically insignificant for the strong growth prospect subsample. These results align with Hypothesis 3, suggesting that when facing higher short-sale pressure due to Reg SHO, pilot firms with poor growth prospects tend to obscure the readability of their 10-Ks as a strategic response.

4.5 Short selling and tone ambiguity

Additionally, ambiguous text in 10-Ks can impede investors' comprehension (Loughran & McDonald, 2011; 2013). In this subsection, we investigate whether pilot firms increase their use of uncertainty words in 10-Ks to hinder short sellers' comprehension. To investigate how changes in short-sale constraints affect the tone ambiguity of 10-Ks, we run the following regression:

$$Uncertain_{i,t} = \alpha + \beta_1 * Pilot_i + \beta_2 * Pilot_i * During_t + \beta_3 * During_t + \gamma \cdot Controls + \varepsilon_{i,t} \quad (2)$$

where $Uncertain_{i,t}$ is the proportion of uncertain words in 10-Ks based on Loughran and McDonald (2011) for firm *i* at year *t*. We employ the set of control variables in Equation (1) to control for the variables that could affect tone ambiguity. We also augment Equation (2) by adding industry and fiscal year fixed effects in column (2) and by adding firm and fiscal year fixed effects

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in column (3). Hypothesis 4 predicts the coefficient of $Pilot_i * During_t$ (β_2) to be significantly positive. Our regression results are reported in Table 6.

< Table 6>

In Table 6, we show that pilot firms significantly increase the proportion of uncertainty words in 10-Ks during the Reg SHO experiment period. The coefficients on $Pilot_i * During_t$ are positive and significant in all columns. The DiD estimator is 0.026 and significant at the 1% level in column (1). This corresponds to approximately 9% (0.026/0.29) of a standard deviation of $Uncertain$ in the full sample. This difference holds economic significance. When examining one of the key drivers of the proportion of uncertain words in 10-Ks, namely, the dummy variable for mergers and acquisitions (MA) events, we observe a 0.015 difference in tone ambiguity between firms with MA activities and those without. This variance accounts for 5% (0.015/0.29) of a standard deviation. After controlling industry and fiscal year dummies, the coefficient $Pilot_i * During_t$ remains positively significant in column (2). $Pilot_i * During_t$ is 0.028 and significant at the 1% level when we add firm and year fixed effects. Our results suggest that pilot firms, whose short-sale constraints are relaxed due to Reg SHO, reduce annual report readability and increase tone ambiguity in 10-Ks. Collectively, the evidence in this subsection further strengthens the incomplete revelation hypothesis in Bloomfield (2002) for explaining the adverse impact of short-selling on the quality of financial disclosure.

5. ROBUSTNESS CHECKS

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In this subsection, we conduct several robustness tests. We perform parallel trend tests and two placebo tests to enhance our causal argument.

5.1 Parallel trend tests

One of the assumptions for difference-in-difference tests is that, in the absence of the treatment, the treated and control groups exhibit parallel trends over time. To prove that the difference observed in previous tests are attributed to Reg SHO, we conduct parallel trend analysis following Wang et al., (2022). We run the regression below:

$$Bog_{i,t} = \alpha + \beta_1 * Pilot_i + \beta_2 * Pilot_i * During_t + \beta_3 * Pilot_i * Pre + Industry_j + Year_t + \varepsilon_{i,t} \quad (3)$$

where *Pre* is a dummy variable that equals one if the end of a firm's fiscal year falls in 2003 and zero otherwise. Our coefficient of interest is the slope of the interaction term between *Pre* and *Pilot*. If the assumption of a parallel trend holds, we expect the coefficient of *Pilot*Pre* to be statistically insignificant. Table 7 reports our results. We also repeat the pre-trend analysis for tone ambiguity by replacing the outcome variable of $Bog_{i,t}$ with $Uncertain_{i,t}$ in Columns (3) and (4).

< Table 7>

Columns (1) and (2) report the results for the impact of Reg SHO on annual report readability, while Columns (3) and (4) describe the results for the effect of Reg SHO on tone ambiguity. We find consistent results with our prediction. In all four columns, the cofficients of *Pilot*Pre* are statistically insignificant, while the coefficients of *Pilot*During* are positive and

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significant. Our findings indicate that during the pre-Reg SHO periods, pilot and non-pilot firms tend to produce 10-Ks with similar readability and tone ambiguity. However, during the Reg SHO periods, pilot firms produce less readable and more ambiguous 10-Ks than non-pilot firms.

5.2 Placebo tests

We perform two sets of placebo tests for our DiD analysis to strengthen our causal argument. We address the concern that our tests mainly depend on the SEC's approval of Reg SHO in 2004. Unobservable shocks that occurred before or after 2004 but are unrelated to Reg SHO could have driven results.

First, we randomly choose 830 stocks to be *pseudo-pilot* stocks in the balanced sample during our event periods. Then we re-estimate our DID regressions for annual report readability and tone ambiguity and report the results in Table 8. As can be seen, the coefficient of interest, *PseudoPilot*During*, is statistically insignificant in each column. In unreported analysis, we drop these pilot stocks and randomly choose 830 stocks to be *pseudo-pilot* stocks. We also obtain insignificant results for *PseudoPilot*During*.

< Table 8>

Second, we use the same pilot and non-pilot firms in Reg SHO but artificially pick two "pseudo-event" years. We assume that Reg SHO is effective either from January 2001 to December 2003 or from January 2011 to December 2013. Next, we re-estimate our DID regressions for annual report readability and tone ambiguity using *PesudoDuring* dummy variables. Table 9 Panels A and B report our regression results for the *pseudo-event* year of 2001 and 2011,

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respectively. As can be seen, the coefficients on $Pilot_i * During_t$ are all insignificant. This result indicates that the identified impacts of short selling on readability and tone ambiguity in 10-Ks are unlikely to be driven by unobserved shocks.

< Table 9>

5.3 Alternative experiment periods and test sample

In our tests, we define during-event periods as between January 2005 and December 2007 using fiscal year ending dates. To confirm that our DiD analysis is robust to alternative pre- and during-event periods, we re-estimate our DiD tests in Tables 3 and 6. In our analysis, we exclude 2004 because the SEC announced the pilot and non-pilot firms in July 2004. We include 2004 in our alternative event periods. We also define test periods using the actual start and end dates of the Reg SHO program. Thus, $During_t$ equals one if a firm's fiscal year end is between May, 2005 and June, 2007 and zero otherwise.

< Table 10>

The regression results for annual report readability and tone ambiguity are presented in Table 10. Panels A and B report the results for the inclusion of 2004 and the experiment periods between May 2005 and June 2007. We find that the coefficients on $Pilot_i * During_t$ are positive and statistically significant in Panels A and B. Our conclusions on the causal effect of short selling on the readability and tone ambiguity of annual reports are unchanged after using alternative during-event periods.

< Table 11>

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Our tests require firms to be in the sample over the pre-event and the during-event periods. We repeat our DID regression analyses for annual report readability and tone ambiguity using the unbalanced sample. Table 11 reports our results. We show that the coefficient on $Pilot_i * During_t$ is significantly positive in each column. Our results further confirm the impact of Reg SHO on the readability and tone ambiguity of annual reports.

6 CONCLUSION

We investigate the causal effects of changes in short-sale constraints on readability and tone ambiguity in the context of annual reports. To establish causality, we employ Reg SHO, which randomly relaxes short-sale constraints for pilot firms during 2005 and 2007. In a DID framework, we find that the relaxation of short-sale constraints reduces annual report readability for pilot firms. The results are more pronounced for firms that receive less investor attention and those with poorer growth prospects. Additionally, we document that pilot firms use higher frequencies of uncertainty words in 10-Ks during the Reg SHO experiment period.

We show that our results are robust to alternative sample periods and regression specifications. Our pre-trend analyses indicate that both pilot and non-pilot firms exhibit similar levels of readability and ambiguity tones in their annual reports prior to Reg SHO. Additionally, we conduct various placebo tests to address the data mining concern. We change the timing of Reg SHO to various periods and repeat the DID regression analyses. Next, we randomly create pilot

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firms without changing the timing of Reg SHO. We find that these fictitious scenarios have no effect on annual report readability and tone ambiguity.

Our findings indicate that the relaxation of short-sale constraints affects corporate managers' financial reporting behavior by producing less readable and more ambiguous 10-Ks. This reporting behavior may decrease the potential amount of short selling when investors have limited attention and processing power. Overall, our study provides important implications to readers of financial disclosures, such as financial analysts and investors.

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Appendix: Definition of Variables

Experiment-related variables:

Pilot: A dummy variable that equals one if a stock is selected as a pilot stock in Regulation SHO's pilot program and zero otherwise.

During: A dummy variable that equals one if the end of a firm's fiscal year *t* falls between May 1, 2005 and June 30, 2007 and zero otherwise.

A firm's annual report readability and ambiguity tone

The Bog Index: Bonsall et al. (2017) construct the Bog Index to measure annual report readability. The index is calculated as follows:

Bog Index = Sentence Bog + Word Bog – Pep

Sentence Bog captures readability issues regarding sentence length. *Word Bog* captures plain English style problems and word difficulty. *Pep* captures good writing attributes that help understanding of texts. An annual report with a higher Bog index is considered less readable.

The proportion of uncertainty words: Loughran and McDonald (2011) develop a list of uncertainty words (e.g. *approximate, contingency, depend, and uncertain*) in financial contexts.

Control variables

Firm size: Larger firms have more complex 10-K reports. The size is defined as the natural logarithm of the market equity of stocks at the end of fiscal year *t*.

Firm age: Older companies have more readable annual reports because there is less information asymmetry and less information uncertainty for these companies. The firm age is the number of years since a company's first appearance in the CRSP monthly stock return file. We use the natural logarithm of the firm age in the regressions.

Special items (SI): Firms with a significant amount of special items are more likely to experience some unusual events. Thus, companies with lower special items have more complex 10-Ks. SI is defined as the amount of special items scaled by book value of assets.

Volatility of business: Stocks with higher volatility have more complex 10-Ks. To capture the volatility of business, we use two measures: stock return volatility (*Ret_vol*, measured as the standard deviation of the monthly stock returns in the prior year) and earnings volatility (*Earn_vol*, measured as the standard deviation of the operating earnings during the prior five fiscal years).

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Profitability (ROA): Firms that earn higher profits have more readable 10-Ks. ROA is defined as the income before extraordinary items divided by lagged total assets.
Complexity of operations: Firms with more complex operations are more likely to have complex 10-Ks. We use the number of business segments (*NBSEG*) and the number of geographic segments (*NGSEG*) to capture the operation complexity of firms. Log(*NBSEG*) is the logarithm of one plus the number of business segments and Log(*NGSEG*) is the logarithm of 1 plus the number of geographic segments.

Financial complexity: Companies with more complex financial situations are more likely to have complicated 10-Ks. We use the logarithm of the number of non-missing items in Compustat as a proxy for financial complexity (*NITEMS*). Firms are more financially complex if they need to report more items in annual reports.

Corporate events: Unusual corporate events may require extra and more detailed disclosures, so firms with corporate events have more complex 10-Ks. We consider two events: seasoned equity offerings (SEOs) and merger and acquisitions (MA). The dummy variable *SEO* is equal to one if for a year in which a company has a common equity offering in the secondary market according to the SDC Global New Issues database and zero otherwise. The dummy variable *MA* is set to one for a year in which a company is an acquirer based on the SDC Platinum M&A database and zero otherwise.

Delaware: Firms incorporated in Delaware have more complex and less readable annual reports. The Delaware dummy variable equals one if a firm is incorporated in Delaware and zero otherwise.

Investor attention proxies

Abnormal trading volume: Firms with higher abnormal trading volume receive more attention. Abnormal trading volume is defined as the ratio of trading volume at the end of each month to the average over the previous one year.

Extreme returns: Firms with higher abnormal extreme returns receive more attention. Extreme returns are defined as the ratio of returns at the end of each month to the average return over the previous one year.

Analyst coverage: Firms that are covered by more analysts receive more investor attention. The analyst coverage is defined as the logarithm of the number of analysts following a stock from IBES database.

Proxies for firms' growth prospects

Cash-flow-to-price ratio: Firms with higher cash-flow-to-price ratios are generally considered to have poorer growth prospects. Cash-flow-to-price is computed as cash flows divided by the market value of equity at the fiscal year-end.

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Earnings-to-price ratio: Firms with higher earnings-to-price ratios are generally considered to have poorer growth prospects. Earnings-to-price ratio is computed as earnings divided by the market value of equity at the fiscal year-end.

Book-to-market ratio: Firms with higher book-to-market ratios are generally considered to have poorer growth prospects. Book-to-market is computed as the book value of common equity scaled by the market value of equity at the fiscal year-end.

Tobin's Q: Firms with lower Tobin's Q are generally considered to have poorer growth prospects. Tobin's Q is defined as total assets plus the market value of equity, minus book value of common equity and minus deferred taxes, scaled by total assets at the fiscal year-end.

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Table 1 Summary statistics

This table reports the summary statistics of firm characteristics of the pilot (treatment) and non-pilot (control) groups measured based on the 2004 Russell 3000 index firms. The sample includes firms whose fiscal year ending dates are between 2001 and 2003 for the pre-event period and between 2005 and 2007 for the during-event period. We require firms to have data available to calculate firm characteristics and bog index over time. Definitions of the variables are provided in the Appendix.

Variables	N	Mean	SD	Minimum	Maximum
Bog Index	11,839	82.95	6.99	66.00	99.00
Uncertainty (%)	11,838	1.37	0.30	0.67	2.10
Log(size)	11,839	7.00	1.57	3.58	11.35
Log(BM)	11,839	-0.82	0.70	-3.03	0.86
Log(age)	11,839	2.82	0.69	0.69	4.42
Log(numbseg)	11,839	1.06	0.44	0.69	2.49
Log(numgseg)	11,839	1.08	0.45	0.69	3.05
EARN_VOL	11,839	0.05	0.07	0.00	0.42
RET_VOL	11,839	0.12	0.08	0.03	0.49
ROA	11,839	0.03	0.13	-0.78	0.29
SI	11,839	-0.01	0.04	-0.34	0.06
Log(nitems)	11,839	5.78	0.16	5.14	6.01
SEO	11,839	0.06	0.24	0.00	1.00
MA	11,839	0.39	0.49	0.00	1.00
Delaware	11,839	0.58	0.49	0.00	1.00

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Table 2 Firm characteristics before announcement of Regulation SHO

This table provides the firm characteristics of the pilot (treatment) and non-pilot (control) groups in 2003. The sample comes from the 2004 Russell 3000 index and contains firms with data available to calculate the Bog Index and control variables. Definitions of variables are in the Appendix. We report the *t*-statistics of the two-sample *t*-test for the difference between the pilot and non-pilot groups. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Treatment			Control			Difference	
	N	Mean	SD	N	Mean	SD	Mean	T-stat
Bog Index	740	82.66	6.45	1463	83.04	6.50	-0.38	-1.30
Uncertainty (%)	740	1.37	0.29	1463	1.39	0.29	-0.02	-1.61
Log(size)	740	7.02	1.39	1463	6.98	1.42	0.04	0.60
Log(BM)	740	-0.88	0.65	1463	-0.91	0.67	0.03	1.11
Log(age)	740	2.79	0.69	1463	2.74	0.69	0.05	1.35
Log(numbseg)	740	1.05	0.43	1463	1.05	0.44	0.00	0.45
Log(numgseg)	740	1.07	0.44	1463	1.07	0.46	0.00	0.01
EARN_VOL	740	0.05	0.07	1463	0.06	0.08	-0.01	-2.23
RET_VOL	740	0.12	0.07	1463	0.12	0.07	0.00	-0.77
ROA	740	0.03	0.11	1463	0.02	0.12	0.01	1.71
SI	740	-0.01	0.03	1463	-0.01	0.03	0.00	-0.63
Log(nitems)	740	5.75	0.16	1463	5.73	0.17	0.02	1.73
SEO	740	0.07	0.25	1463	0.08	0.27	-0.01	-0.65
MA	740	0.40	0.49	1463	0.36	0.48	0.04	1.67
Delaware	740	0.60	0.49	1463	0.59	0.49	0.01	0.52

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Table 3 Multivariate difference-in-differences tests: annual report readability and Regulation SHO

This table presents the results of DiD tests examining how the relaxation of short-sale constraints affects annual report readability using a balanced panel. The sample comes from the 2004 Russell 3000 index and contains firms that have data available to calculate readability and control variables over the pre-event (fiscal year ending date is between 2001 and 2003) and during-event (fiscal year ending date is between 2005 and 2007) periods. Column (1) reports the results of the following regression:

$$Bog\ Index_{i,t} = \alpha + \beta_1 * pilot_i + \beta_2 * pilot_i * during_t + Industry_j + Year_t + \varepsilon_{i,t,}$$

where $Bog\ Index_{i,t}$ is the Bog Index (Bonsall et al. (2017)) for firm *i* at year *t*. $Pilot_i$ is a dummy variable that equals one if a stock is selected as a pilot stock in Regulation SHO's pilot program and zero otherwise. $During_t$ is a dummy variable that equals one if the end of a firm's fiscal year *t* falls between 2005 and 2007 and zero otherwise. *Industry* and *Year* are the industry fixed effects (3-digit SIC codes) and fiscal year fixed effects, respectively. We omit $during_t$ to avoid multicollinearity. We add control variables to the regression and provide the results without fixed effects in column (1), with industry and year fixed effects in column (2), and with firm and year fixed effects in column (3). Variable definitions are provided in the Appendix. Standard errors clustered by firms are displayed in parentheses. ***, ** and * indicate significance at the 1, 5 and 10 percent levels, respectively.

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Variables	(1)	(2)	(3)
Pilot	-0.386	-0.404	
	(0.293)	(0.253)	
*Pilot*During*	0.460**	0.458**	0.508***
	(0.223)	(0.204)	(0.185)
During	2.373***		
	(0.210)		
Log(Size)	0.511***	0.544***	0.157
	(0.094)	(0.085)	(0.129)
Log(BM)	0.067	0.646***	0.407***
	(0.174)	(0.149)	(0.131)
Log(age)	-0.792***	-0.592***	-1.607***
	(0.188)	(0.175)	(0.563)
Log(numbseg)	1.509***	1.431***	0.791***
	(0.277)	(0.256)	(0.228)
Log(numgseg)	1.285***	-0.102	0.175
	(0.259)	(0.267)	(0.254)
Earn_vol	7.008***	0.582	0.915
	(1.803)	(1.495)	(1.158)
Ret_vol	11.640***	7.533***	2.429***
	(1.286)	(1.127)	(0.725)
ROA	-11.691***	-5.327***	-1.928***
	(1.024)	(0.884)	(0.608)
SI	3.345	0.128	-0.355
	(2.061)	(1.728)	(1.129)
Log(nitems)	7.672***	1.183	1.358
	(0.799)	(1.613)	(1.061)
SEO	0.068	-0.337	-0.144
	(0.265)	(0.223)	(0.138)
MA	0.705***	0.441***	0.239***
	(0.155)	(0.125)	(0.076)
Delaware	0.105	0.299	
	(0.250)	(0.222)	
Industry FE	NO	YES	NO
Firm FE	NO	NO	YES
Year FE	NO	YES	YES
Observations	11,839	11,836	11,722
Adjusted R-squared	0.218	0.463	0.828

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Table 4 Annual report readability and Regulation SHO: sample partitioned by investor attention

This table describes how investor attention impacts the effect of short-selling on annual report readability. We use three measures of investor attention: (1) abnormal trading volume (Barber & Odean, 2008); (2) extreme returns (Barber & Odean, 2008); (3) analyst coverage (Hirshleifer & Teoh, 2003). For each proxy, we partition the sample into high and low investor attention subsamples based on its median values each year. Then we repeat DiD tests in Table 3 across each subsample. We present the results with firm and year fixed effects. Variable definitions are provided in the Appendix. Standard errors clustered by firms are displayed in parentheses. ***, ** and * indicate statistical significance at the 1, 5 and 10 percent levels, respectively.

	Abnormal Trading Volume		Extreme Returns		Analyst Coverage	
	Low	High	Low	High	Low	High
Variables	(1)	(2)	(3)	(4)	(5)	(6)
*Pilot*During*	0.505**	0.292	0.719***	0.437*	0.804***	0.281
	(0.245)	(0.255)	(0.244)	(0.251)	(0.280)	(0.288)
Log(Size)	0.212	0.191	0.257	-0.040	0.144	0.016
	(0.184)	(0.189)	(0.192)	(0.182)	(0.189)	(0.228)
Log(BM)	0.476***	0.479**	0.525***	0.341*	0.396**	0.506**
	(0.177)	(0.222)	(0.185)	(0.202)	(0.198)	(0.217)
Log(age)	-1.245	-1.669**	-1.008	-1.752**	-1.711**	-1.669*
	(0.808)	(0.751)	(0.752)	(0.787)	(0.868)	(0.955)
Log(numbseg)	1.024***	0.780**	0.715**	0.910***	0.566*	0.860**
	(0.332)	(0.335)	(0.330)	(0.333)	(0.326)	(0.335)
Log(numgseg)	-0.519	0.277	-0.030	0.298	0.383	-0.163
	(0.382)	(0.362)	(0.384)	(0.342)	(0.344)	(0.395)
Earn_vol	0.134	2.402	0.334	1.122	1.149	0.539
	(1.673)	(1.685)	(1.659)	(1.831)	(1.559)	(2.196)
Ret_vol	1.714*	3.407***	1.765*	3.314***	3.083***	-0.166
	(1.006)	(1.239)	(1.025)	(1.149)	(1.035)	(1.174)
ROA	-1.764**	-2.115**	-2.856***	-0.852	-1.478*	-1.240
	(0.839)	(1.038)	(0.889)	(0.852)	(0.787)	(1.115)
SI	-0.765	-0.202	2.120	-2.822*	-0.742	-2.678
	(1.688)	(1.822)	(1.735)	(1.648)	(1.464)	(2.033)
Log(nitems)	4.357***	-0.917	2.250	1.757	1.657	-0.809
	(1.573)	(1.501)	(1.415)	(1.397)	(1.602)	(1.680)
SEO	-0.011	-0.173	0.045	-0.306	-0.331*	0.057
	(0.254)	(0.206)	(0.240)	(0.225)	(0.192)	(0.215)
MA	0.188	0.305***	0.287**	0.044	0.345***	0.185
	(0.122)	(0.117)	(0.119)	(0.120)	(0.117)	(0.116)
Firm FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	5,307	5,470	5,406	5,397	5,451	4,597
Adjusted R-squared	0.830	0.817	0.833	0.822	0.825	0.839

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Table 5 Annual report readability and Regulation SHO: sample partitioned by firms' growth prospects

This table examines how firms' growth prospects influence the effect of short-selling on annual report readability. We employ four measures of firms' growth prospects: fundamentals-to-market ratios, which include the cash-flow-to-price ratio, earnings-to-price ratio, and book-to-market ratio (Dechow et al., 2001); and Tobin's Q (Lang, Ofek, & Stulz, 1996). Firms with high fundamentals-to-market ratios or low Tobin's Q exihibit poor growth prospects. We partition the samples into subgroups of poor and strong prospects based on the median value of each proxy every year. Subsequently, we perform DiD tests in Table 3 for each subsample. The results include firm and year fixed effects, with variable definitions available in the Appendix. Standard errors clustered by firms are displayed in parentheses. ***, ** and * indicate statistical significance at the 1, 5 and 10 percent levels, respectively.

	Cash-flow-to-price		Earnings-to-price		Book-to-market		Tobin's Q	
	Poor Prospect	Strong Prospect	Poor Prospect	Strong Prospect	Poor Prospect	Strong Prospect	Poor Prospect	Strong Prospect
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
*Pilot*During*	0.565**	0.309	0.794***	0.274	0.621**	0.266	0.759***	0.204
	(0.241)	(0.295)	(0.253)	(0.292)	(0.245)	(0.299)	(0.243)	(0.299)
Log(Size)	0.179	0.060	0.191	0.161	-0.069	0.205	0.229	-0.015
	(0.195)	(0.184)	(0.146)	(0.296)	(0.145)	(0.194)	(0.181)	(0.214)
Log(BM)	0.514**	0.216	0.294**	0.378			0.379**	0.379*
	(0.200)	(0.185)	(0.146)	(0.306)			(0.172)	(0.211)
Log(age)	-1.567**	-1.151	-0.550	-1.559*	-1.470**	-1.514	-1.930**	-0.552
	(0.747)	(0.932)	(0.771)	(0.926)	(0.739)	(0.979)	(0.760)	(0.935)
Log(numbseg)	0.932***	0.536*	0.942***	0.658**	1.304***	0.672**	0.748**	0.727*
	(0.342)	(0.295)	(0.328)	(0.331)	(0.345)	(0.321)	(0.290)	(0.373)
Log(numgseg)	0.258	0.295	0.212	-0.019	0.114	0.136	0.450	-0.148
	(0.388)	(0.376)	(0.309)	(0.441)	(0.318)	(0.391)	(0.348)	(0.349)
Earn_vol	-0.425	1.033	-0.086	4.736**	1.625	0.369	1.727	2.702*
	(1.969)	(1.510)	(1.431)	(2.250)	(1.445)	(2.137)	(1.810)	(1.475)
Ret_vol	3.217**	1.642*	2.861***	2.535	2.135**	2.868**	3.166***	0.902
	(1.265)	(0.963)	(0.846)	(1.744)	(0.923)	(1.241)	(0.972)	(1.222)
ROA	-3.052***	-1.374*	-0.983	-5.151**	-1.210	-4.536***	-1.892**	-1.633
	(1.143)	(0.768)	(0.720)	(1.997)	(0.766)	(1.207)	(0.881)	(1.003)
SI	0.029	-1.546	-0.788	0.783	-1.212	3.432	-0.579	-2.047
	(2.057)	(1.439)	(1.225)	(4.590)	(1.397)	(2.166)	(1.560)	(1.853)
Log(nitems)	1.159	-0.626	1.668	4.397	0.906	0.959	1.199	-0.853
	(1.221)	(2.443)	(1.224)	(2.694)	(1.195)	(2.614)	(1.218)	(2.463)
SEO	0.045	-0.172	-0.196	0.050	-0.089	-0.049	-0.303	0.009
	(0.193)	(0.226)	(0.182)	(0.238)	(0.177)	(0.225)	(0.199)	(0.232)
MA	0.276***	0.250**	0.153	0.250**	0.181*	0.174	0.161	0.283**
	(0.101)	(0.127)	(0.101)	(0.124)	(0.099)	(0.127)	(0.102)	(0.125)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Observations	7,054	4,155	6,545	4,619	7,109	4,419	7,083	4,148
Adjusted R-squared	0.811	0.843	0.841	0.821	0.839	0.816	0.819	0.856

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Table 6 Multivariate difference-in-differences tests: tone ambiguity and Regulation SHO

This table provides the results of multivariate DiD tests on the impact of the relaxation of short-sale constraints on tone ambiguity in annual reports. We run the following OLS regressions:

$$Uncertain_{i,t} = \alpha + \beta_1 * pilot_i + \beta_2 * pilot_i * during_t + Industry_j + Year_t + \varepsilon_{i,t}$$

where $Uncertain_{i,t}$ is the proportion of uncertain words in 10-Ks based on Loughran and McDonald (2011) for firm *i* at year *t*. $Pilot_i$ is a dummy variable that equals one if a stock is selected as a pilot stock in Regulation SHO's pilot program and zero otherwise. $During_t$ is a dummy variable that equals one if the end of a firm's fiscal year *t* falls between 2005 and 2007 and zero otherwise. *Industry* and *Year* are the industry fixed effects (2 digits SIC codes) and fiscal year fixed effects, respectively. We replace the industry fixed effect with the firm fixed effect in column (3). The sample comes from the 2004 Russell 3000 index and contains firms that have data available to calculate $Uncertain_{i,t}$ and controls over the pre-event (fiscal year ending date is between 2001 and 2003) and during-event periods. Variable definitions are provided in the Appendix. Standard errors clustered by firms are displayed in parentheses. ***, ** and * indicate statistical significance at the 1, 5 and 10 percent levels, respectively.

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Variables	(1)	(2)	(3)
Pilot	-0.021*	-0.015	
	(0.011)	(0.011)	
*Pilot*During*	0.026***	0.021**	0.028***
	(0.010)	(0.010)	(0.009)
During	0.183***		
	(0.009)		
Log(Size)	0.016***	0.017***	0.037***
	(0.004)	(0.004)	(0.006)
Log(BM)	0.016**	0.024***	0.017**
	(0.006)	(0.006)	(0.007)
Log(age)	-0.084***	-0.081***	-0.206***
	(0.007)	(0.007)	(0.029)
Log(numbseg)	-0.069***	-0.060***	-0.019
	(0.010)	(0.011)	(0.012)
Log(numgseg)	0.031***	-0.008	0.037***
	(0.010)	(0.011)	(0.014)
Earn_vol	0.438***	0.258***	0.032
	(0.068)	(0.069)	(0.070)
Ret_vol	0.469***	0.466***	0.200***
	(0.052)	(0.049)	(0.038)
ROA	-0.114***	-0.006	-0.082**
	(0.039)	(0.041)	(0.034)
SI	0.018	0.007	0.037
	(0.085)	(0.083)	(0.063)
Log(nitems)	0.222***	-0.026	-0.030
	(0.033)	(0.065)	(0.054)
SEO	0.033***	0.033***	0.012
	(0.011)	(0.010)	(0.008)
MA	0.015**	0.011**	0.010**
	(0.006)	(0.006)	(0.004)
Delaware	0.027***	0.030***	
	(0.009)	(0.009)	
Industry FE	NO	YES	NO
Firm FE	NO	NO	YES
Year FE	NO	YES	YES
Observations	12,241	12,238	12,149
Adjusted R-squared	0.243	0.366	0.708

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Table 7 Parallel trend tests: readability and tone ambiguity

This table reports the results of multivariate DiD test examining the parallel trend assumption. Column (1) reports the results of the following regression:

$$Bog_{i,t} = \alpha + \beta_1 * pilot_i + \beta_2 * pilot_i * during_t + \beta_3 * pilot_i * pre + Industry_j + Year_t + \varepsilon_{i,t}$$

where *Pre* is a dummy variable that equals one if the end of a firm's fiscal year falls in 2003 and zero otherwise. We present the results with firm and year fixed effects in Column (2). In Columns (3) and (4), we present the results with tone ambiguity. Variable definitions are provided in the Appendix. Standard errors clustered by firms are displayed in parentheses. ***, ** and * indicate statistical significance at the 1, 5 and 10 percent levels, respectively.

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	Readability		Tone Ambiguity	
Variables	(1)	(2)	(3)	(4)
Pilot	-0.410		-0.019	
	(0.275)		(0.012)	
*Pilot*During*	0.463**	0.561***	0.024**	0.033***
	(0.233)	(0.214)	(0.011)	(0.011)
*Pilot*Pre*	0.005	0.150	0.011	0.013
	(0.185)	(0.168)	(0.010)	(0.009)
Log(Size)	0.533***	0.159	0.017***	0.037***
	(0.085)	(0.129)	(0.004)	(0.006)
Log(BM)	0.651***	0.422***	0.024***	0.016**
	(0.149)	(0.131)	(0.006)	(0.007)
Log(age)	-0.575***	-1.617***	-0.080***	-0.206***
	(0.176)	(0.568)	(0.007)	(0.029)
Log(numbseg)	1.431***	0.781***	-0.060***	-0.020
	(0.257)	(0.228)	(0.011)	(0.012)
Log(numgseg)	-0.117	0.164	-0.009	0.036**
	(0.268)	(0.254)	(0.011)	(0.014)
Earn_vol	0.493	0.802	0.248***	0.020
	(1.509)	(1.163)	(0.069)	(0.070)
Ret_vol	7.602***	2.534***	0.491***	0.209***
	(1.136)	(0.729)	(0.049)	(0.038)
ROA	-5.274***	-1.910***	0.002	-0.079**
	(0.884)	(0.613)	(0.041)	(0.034)
SI	0.099	-0.415	-0.003	0.029
	(1.742)	(1.131)	(0.083)	(0.063)
Log (nitems)	1.932	1.842*	-0.040	-0.026
	(1.599)	(1.065)	(0.065)	(0.054)
SEO	-0.382*	-0.148	0.031***	0.012
	(0.224)	(0.138)	(0.010)	(0.008)
MA	0.458***	0.242***	0.012**	0.010**
	(0.126)	(0.076)	(0.006)	(0.004)
Delaware	0.275		0.028***	
	(0.223)		(0.009)	
Industry FE	YES	NO	YES	NO
Firm FE	NO	YES	NO	YES
Year FE	YES	YES	YES	YES
Observations	11,836	11,722	12,238	12,149
Adjusted R-squared	0.461	0.828	0.369	0.709

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Table 8 Placebo tests using pseudo pilot firms

This table reports the multivariate DiD test results on how the relaxation of short-sale constraints affects annual report readability and tone ambiguity using pseudo-pilot stocks. We randomly choose 830 stocks to be pilot stocks during our event periods in our balanced sample. Then we re-estimate DID regressions using pseudo pilot stocks. Variable definitions are provided in the Appendix. Standard errors clustered by firms are in parentheses. ***, ** and * indicate statistical significance at the 1, 5 and 10 percent levels, respectively.

	Readability		Tone Ambiguity	
Variables	(1)	(2)	(3)	(4)
PseudoPilot	0.337		0.013	
	(0.256)		(0.011)	
*PseudoPilot*During*	-0.083	0.113	0.001	0.002
	(0.208)	(0.190)	(0.010)	(0.010)
Log(Size)	0.541***	0.166	0.017***	0.037***
	(0.085)	(0.129)	(0.004)	(0.007)
Log(BM)	0.636***	0.420***	0.023***	0.017**
	(0.149)	(0.131)	(0.006)	(0.007)
Log(age)	-0.593***	-1.654***	-0.082***	-0.212***
	(0.175)	(0.563)	(0.007)	(0.029)
Log(numbseg)	1.439***	0.801***	-0.060***	-0.020
	(0.256)	(0.228)	(0.011)	(0.012)
Log(numgseg)	-0.113	0.200	-0.009	0.035**
	(0.268)	(0.255)	(0.012)	(0.014)
Earn_vol	0.687	1.060	0.260***	0.045
	(1.497)	(1.142)	(0.069)	(0.071)
Ret_vol	7.569***	2.491***	0.470***	0.202***
	(1.128)	(0.729)	(0.049)	(0.038)
ROA	-5.369***	-1.955***	-0.014	-0.085**
	(0.889)	(0.609)	(0.041)	(0.035)
SI	0.090	-0.409	0.011	0.034
	(1.734)	(1.133)	(0.083)	(0.064)
Log (nitems)	1.273	1.234	-0.025	-0.044
	(1.616)	(1.062)	(0.066)	(0.055)
SEO	-0.335	-0.133	0.038***	0.017**
	(0.224)	(0.140)	(0.010)	(0.008)
MA	0.438***	0.234***	0.010*	0.009**
	(0.126)	(0.076)	(0.006)	(0.004)
Delaware	0.294		0.031***	
	(0.222)		(0.010)	
Industry FE	YES	NO	YES	NO
Firm FE	NO	YES	NO	YES
Year FE	YES	YES	YES	YES
Observations	11,802	11,701	12,054	11,966
Adjusted R-squared	0.463	0.828	0.368	0.707

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Table 9 Placebo tests using pseudo event periods

This table reports the multivariate DiD test results on how the relaxation of short-sale constraints affects annual report readability and tone ambiguity using two pseudo experiment periods. We assume that Reg SHO starts from January 2001 or January 2011. Then we repeat the DID analysis with *PseudoDuring* dummy variable. Variable definitions are provided in the Appendix. Standard errors clustered by firms are in parentheses. ***, ** and * indicate statistical significance at the 1, 5 and 10 percent levels, respectively.

Panel A: Pseudo experiment periods between 2001 and 2003				
	Readability		Tone Ambiguity	
Variables	(1)	(2)	(3)	(4)
Pilot	-0.223		-0.011	
	(0.309)		(0.012)	
*Pilot*PsuedoDuring*	-0.161	-0.049	-0.002	-0.003
	(0.225)	(0.194)	(0.010)	(0.009)
Log(Size)	0.513***	-0.179	0.019***	0.030***
	(0.103)	(0.141)	(0.004)	(0.006)
Log(BM)	0.682***	0.328**	0.016**	0.019***
	(0.163)	(0.143)	(0.007)	(0.006)
Log(age)	-1.281***	-5.883***	-0.090***	-0.035
	(0.220)	(0.712)	(0.009)	(0.030)
Log(numbseg)	1.578***	1.113***	-0.041***	0.003
	(0.296)	(0.224)	(0.012)	(0.009)
Log(numgseg)	-0.269	-0.078	-0.004	0.026**
	(0.334)	(0.266)	(0.015)	(0.013)
Earn_vol	5.326**	0.460	0.428***	0.188*
	(2.280)	(2.131)	(0.081)	(0.108)
Ret_vol	6.322***	-0.879	0.457***	0.144***
	(1.208)	(0.732)	(0.051)	(0.035)
ROA	-5.425***	-2.366***	0.039	-0.030
	(0.969)	(0.771)	(0.045)	(0.037)
SI	0.401	0.385	-0.103	0.017
	(1.838)	(1.239)	(0.090)	(0.065)
Log (nitems)	3.293	-2.138	0.141	0.081
	(2.499)	(1.623)	(0.097)	(0.075)
SEO	-0.163	-0.070	0.036***	-0.000
	(0.249)	(0.163)	(0.011)	(0.008)
MA	0.504***	0.012	0.005	0.001
	(0.152)	(0.089)	(0.007)	(0.004)
Delaware	0.442		0.035***	
	(0.277)		(0.011)	
Industry FE	YES	NO	YES	NO
Firm FE	NO	YES	NO	YES
Year FE	YES	YES	YES	YES
Observations	10,184	10,027	10,533	10,400
Adjusted R-squared	0.451	0.829	0.332	0.728

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Panel B: Pseudo experiment periods between 2011 and 2013				
	Readability		Tone Ambiguity	
Variables	(1)	(2)	(3)	(4)
Pilot	0.072		-0.004	
	(0.247)		(0.011)	
*Pilot*PsuedoDuring*	-0.348**	-0.182	-0.003	-0.005
	(0.144)	(0.130)	(0.007)	(0.006)
Log(Size)	0.469***	0.004	0.003	0.015***
	(0.094)	(0.105)	(0.004)	(0.005)
Log(BM)	0.623***	0.362***	0.007	0.013***
	(0.159)	(0.097)	(0.007)	(0.005)
Log(age)	-0.180	-0.111	-0.066***	-0.104***
	(0.223)	(0.741)	(0.011)	(0.035)
Log(numbseg)	1.210***	0.785***	-0.063***	-0.022**
	(0.273)	(0.260)	(0.013)	(0.011)
Log(numgseg)	-0.376	-0.042	-0.002	0.014
	(0.298)	(0.234)	(0.013)	(0.012)
Earn_vol	1.967	1.596	0.058	-0.126*
	(2.495)	(1.536)	(0.106)	(0.068)
Ret_vol	6.327***	1.311***	0.014	-0.015
	(1.149)	(0.493)	(0.052)	(0.026)
ROA	-6.149***	-1.303***	-0.002	-0.006
	(1.033)	(0.436)	(0.042)	(0.022)
SI	0.821	-0.207	-0.072	-0.059
	(1.597)	(0.648)	(0.067)	(0.037)
Log (nitems)	4.758	4.264***	0.226*	-0.002
	(2.943)	(1.416)	(0.119)	(0.064)
SEO	0.542**	0.148	0.036***	0.012**
	(0.227)	(0.102)	(0.010)	(0.006)
MA	0.511***	0.148***	0.007	0.009***
	(0.131)	(0.055)	(0.006)	(0.003)
Delaware	0.204		0.021*	
	(0.242)		(0.012)	
Industry FE	YES	NO	YES	NO
Firm FE	NO	YES	NO	YES
Year FE	YES	YES	YES	YES
Observations	9,906	9,874	10,198	10,189
Adjusted R-squared	0.413	0.886	0.264	0.814

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Table 10 Robustness: alternative experiment periods

This table reports the multivariate DiD test results on how the relaxation of short-sale constraints affects annual report readability and tone ambiguity using alternative experiment periods. First, we add the 2004 sample to our pre-event sample and then re-estimate our DID regressions. Second, we define the during-event period as from May 2005 to June 2007 and then repeat our DID tests to align closely with the actual effective period of Reg SHO. Variable definitions are provided in the Appendix. Standard errors clustered by firms are in parentheses. ***, ** and * indicate statistical significance at the 1, 5 and 10 percent levels, respectively.

Panel A: The inclusion of 2004				
	Readability		Tone Ambiguity	
Variables	(1)	(2)	(3)	(4)
Pilot	-0.399*		-0.015	
	(0.235)		(0.010)	
*Pilot*During*	0.488***	0.506***	0.021**	0.022***
	(0.171)	(0.150)	(0.008)	(0.008)
Log(Size)	0.562***	0.194	0.017***	0.037***
	(0.084)	(0.120)	(0.004)	(0.006)
Log(BM)	0.716***	0.426***	0.024***	0.019***
	(0.147)	(0.118)	(0.006)	(0.006)
Log(age)	-0.544***	-1.543***	-0.080***	-0.194***
	(0.169)	(0.533)	(0.007)	(0.027)
Log(numbseg)	1.403***	0.784***	-0.061***	-0.017
	(0.256)	(0.209)	(0.011)	(0.012)
Log(numgseg)	-0.069	0.145	-0.010	0.036***
	(0.268)	(0.227)	(0.011)	(0.013)
Earn_vol	0.356	1.201	0.272***	0.038
	(1.474)	(1.077)	(0.069)	(0.065)
Ret_vol	7.680***	2.365***	0.437***	0.184***
	(1.107)	(0.641)	(0.048)	(0.034)
ROA	-5.304***	-1.995***	0.002	-0.081***
	(0.883)	(0.550)	(0.041)	(0.031)
SI	-0.081	-0.266	0.030	0.055
	(1.711)	(1.024)	(0.082)	(0.059)
Log (nitems)	0.729	1.027	0.018	-0.052
	(1.465)	(0.831)	(0.059)	(0.044)
SEO	-0.294	-0.133	0.036***	0.015**
	(0.202)	(0.121)	(0.010)	(0.007)
MA	0.413***	0.256***	0.013**	0.010***
	(0.118)	(0.067)	(0.005)	(0.004)
Delaware	0.284		0.030***	
	(0.220)		(0.009)	
Industry FE	YES	NO	YES	NO
Firm FE	NO	YES	NO	YES
Year FE	YES	YES	YES	YES
Observations	14,007	13,912	14,484	14,411
Adjusted R-squared	0.457	0.835	0.357	0.719

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Panel B: Experiment periods between May 2005 and June 2007				
	Readability		Tone Ambiguity	
Variables	(1)	(2)	(3)	(4)
Pilot	-0.305		-0.015	
	(0.229)		(0.010)	
*Pilot*During*	0.458***	0.373***	0.026***	0.025***
	(0.164)	(0.142)	(0.009)	(0.008)
Log(Size)	0.522***	0.140	0.018***	0.041***
	(0.086)	(0.122)	(0.004)	(0.007)
Log(BM)	0.698***	0.510***	0.025***	0.022***
	(0.157)	(0.120)	(0.007)	(0.007)
Log(age)	-0.538***	-1.417***	-0.084***	-0.180***
	(0.170)	(0.536)	(0.007)	(0.032)
Log(numbseg)	1.459***	0.762***	-0.058***	-0.018
	(0.262)	(0.213)	(0.011)	(0.013)
Log(numgseg)	-0.023	0.489**	-0.011	0.031**
	(0.275)	(0.216)	(0.012)	(0.014)
Earn_vol	-0.039	1.258	0.243***	0.031
	(1.496)	(1.061)	(0.071)	(0.067)
Ret_vol	7.129***	1.723**	0.449***	0.210***
	(1.245)	(0.678)	(0.056)	(0.038)
ROA	-5.104***	-1.742***	-0.003	-0.101***
	(0.930)	(0.540)	(0.042)	(0.029)
SI	0.412	-0.795	0.069	0.065
	(1.789)	(1.003)	(0.084)	(0.058)
Log (nitems)	1.222	1.162	0.051	-0.057
	(1.462)	(0.814)	(0.060)	(0.046)
SEO	-0.238	-0.040	0.035***	0.013*
	(0.207)	(0.121)	(0.010)	(0.007)
MA	0.498***	0.248***	0.015***	0.012***
	(0.122)	(0.067)	(0.006)	(0.004)
Delaware	0.327		0.032***	
	(0.221)		(0.010)	
Industry FE	YES	NO	YES	NO
Firm FE	NO	YES	NO	YES
Year FE	YES	YES	YES	YES
Observations	12,036	11,960	12,443	12,382
Adjusted R-squared	0.446	0.844	0.330	0.718

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Table 11 Robustness: unbalanced Sample

This table reports the multivariate DiD test results on how the relaxation of short-sale constraints affects annual report readability and tone ambiguity using the unbalanced sample. The sample contains firms with data available to calculate readability and control variables over the pre-event (fiscal year ending date is between 2001 and 2003) and during-event (fiscal year ending date is between 2005 and 2007) periods. Standard errors clustered by firms are in parentheses. ***, ** and * indicate statistical significance at the 1, 5 and 10 percent levels, respectively.

	Readability		Tone Ambiguity	
Variables	(1)	(2)	(3)	(4)
Pilot	-0.419*		-0.013	
	(0.246)		(0.011)	
*Pilot*During*	0.503**	0.517***	0.017*	0.029***
	(0.202)	(0.183)	(0.010)	(0.009)
Log(Size)	0.526***	0.161	0.017***	0.035***
	(0.082)	(0.123)	(0.004)	(0.006)
Log(BM)	0.576***	0.382***	0.020***	0.013**
	(0.130)	(0.112)	(0.006)	(0.006)
Log(age)	-0.563***	-1.589***	-0.077***	-0.197***
	(0.166)	(0.548)	(0.007)	(0.028)
Log(numbseg)	1.422***	0.791***	-0.062***	-0.018
	(0.251)	(0.225)	(0.010)	(0.012)
Log(numgseg)	-0.117	0.188	-0.011	0.034**
	(0.262)	(0.254)	(0.011)	(0.014)
Earn_vol	0.215	0.839	0.252***	0.032
	(1.466)	(1.143)	(0.067)	(0.070)
Ret_vol	7.391***	2.303***	0.483***	0.201***
	(1.106)	(0.709)	(0.048)	(0.037)
ROA	-5.411***	-1.981***	0.008	-0.077**
	(0.864)	(0.602)	(0.041)	(0.034)
SI	0.495	-0.169	-0.009	0.037
	(1.652)	(1.084)	(0.081)	(0.061)
Log (nitems)	1.127	1.376	-0.013	-0.031
	(1.570)	(1.042)	(0.064)	(0.054)
SEO	-0.325	-0.125	0.035***	0.013*
	(0.218)	(0.134)	(0.010)	(0.008)
MA	0.453***	0.225***	0.012**	0.010**
	(0.123)	(0.075)	(0.006)	(0.004)
Delaware	0.329		0.030***	
	(0.217)		(0.009)	
Industry FE	YES	NO	YES	NO
Firm FE	NO	YES	NO	YES
Year FE	YES	YES	YES	YES
Observations	12,151	11,996	12,577	12,446
Adjusted R-squared	0.461	0.830	0.362	0.711

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